# Koushik Sivarama Krishnan

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LinkedIn Github Portfolio

## Highlights

- Master's in Applied Data Science with extensive experience building end-to-end AI systems across **LLMs**, **NLP**, **RAG**, and production ML, including deployed internal assistants and evaluation pipelines for real-world users.
- Partnered with claims adjusters, researchers, and cross-functional teams to translate ambiguous business needs into deployable AI solutions that reduced manual effort, accelerated decision support, and improved the usefulness of AI outputs for end users.
- Specialized in **Multi-agent Workflows**, and model evaluation; building reliability-focused AI systems with regression testing, tool integration, and production deployments that improved trustworthiness, consistency, and operational readiness.

## Skills

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| **Programming & Backend** | Python, SQL, Bash, FastAPI, Flask, REST APIs, PostgreSQL |
| **LLMs, RAG & Agents** | LangChain, LangGraph, CrewAI, Retrieval Augmented Generation (RAG), embeddings, vector search, Model Context Protocol (MCP), OpenAI API, HuggingFace, Prompt/Context Engineering |
| **MLOps & Quality** | evaluation pipelines, model benchmarking, MLflow, Weights & Biases, DVC, GitHub Actions CI, tox-based unit tests, Grafana, Prometheus |
| **Data / Analytics** | Pandas, NumPy, PySpark, Power BI, SHAP, Topic Modeling, Data Analysis |
| **Cloud / Infrastructure** | Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, GPU Clusters, Distributed Computing, Truefoundry |

## Experience

### Co-op Data Scientist — *09/2024 - 08/2025*

Insurance Corporation of British Columbia (ICBC) — *Vancouver, BC*

- Scoped and validated high-impact internal GenAI use cases by working with business stakeholders to identify knowledge-access bottlenecks, assess feasibility, and prioritize workflows where AI-assisted retrieval could reduce manual search effort.
- Architected and deployed ICBC's in-house agentic GenAI assistant using **LangChain RAG**, **Milvus**, and **MCP** tools on **Azure Kubernetes Service (AKS)**, building modular retrievers, tool interfaces, and prompt-aware components to support multi-step workflows.
- Designed an automated **LLM evaluation** and regression-testing pipeline across accuracy, faithfulness, consistency, and bias, enabling safer validation of prompt, retriever, and model changes before production release.
- Reduced internal knowledge retrieval time by ~**60%**, decreasing manual lookup effort for claims adjusters and improving customer experience by shortening hold times during claims support interactions.

### Research Engine Developer — *05/2024 - 02/2025*

Justice Data and Design Lab — Victoria, BC

- Tackled access-to-justice challenges for British Columbians by improving a legal aid QA assistant that helped people find relevant legal information and navigate to appropriate legal resources more efficiently.
- Redesigned the **RAG-based legal question-answering pipeline** with cleaner text preprocessing, improved chunking, and **LLM-based reranking** to increase relevant document retrieval quality and strengthen response grounding.
- **Doubled** downstream QA chatbot effectiveness by improving relevant document retrieval accuracy by **2x**, while delivering Power BI-based legal text mining insights that informed outreach and intake workflow improvements for partner organizations.

### Founding Machine Learning Engineer — *04/2023 - 07/2023*

SeiSei.ai — *India*

- Built and deployed an end-to-end Hindi voice conversion pipeline, improving FreeVC performance by **15%** through model benchmarking, hyperparameter tuning, and custom audio preprocessing.
- Architected an automated audio pipeline processing **5,000+ hours** of data, covering collection, preprocessing, quality checks, and batch inference to support reliable model iteration and deployment.
- Led the company's early ML execution by recruiting and mentoring interns, coordinating stakeholder feedback loops, and deploying models through **Kubernetes** on Truefoundry with monitoring and fail-safe evaluation practices.

**Computer Vision Intern** *09/2021 – 02/2022*

Drive Analytics *India*

- Supported production sports analytics workflows by helping design systems that could process large volumes of game footage reliably under strict delivery timelines for external league partners.
- Built and deployed production video analytics pipelines on **AWS** using distributed workers with **Celery** and **RabbitMQ**, processing ∼**10K clips/day** and returning analytics within a **2-hour** delivery window.
- Strengthened downstream trust in generated analytics by iterating on failure cases across multi-view footage, low-resolution clips, occlusions, and weather noise, reducing misleading outputs in production use.

**Deep Learning Intern** *12/2020 – 06/2021*

MURF.ai India

- Contributed to speech AI development by building systems that improved the quality and robustness of voice synthesis and transcription workflows across diverse audio inputs.
- Developed and deployed a voice cloning application using **FastSpeech-2**, building advanced audio preprocessing pipelines with librosa and custom feature engineering for high-quality speech synthesis.
- Fine-tuned speech-to-text models in **Azure Speech Studio**, improving transcription quality through acoustic feature optimization and domain-focused preprocessing.

## Education

**Master of Engineering in Applied Data Science,** *University of Victoria* **Victoria, BC** *2023-2025*

Notable Coursework: Optimization for Machine Learning, Scaling Large Language Models on GPUs, Data Privacy

**Bachelor of Engineering in Computer Science and Engineering,** *Anna University* **India** *2019-2023*

Notable Coursework: Artificial Intelligence, Cloud Computing, Data Structures, Data Warehousing, and Data Mining

## Projects

**GenZ AI Therapist** *02/2026*

- Created a **personalized AI wellness platform** for non-clinical support, enabling users to maintain ongoing conversations, journal experiences, and track emotional wellness signals over time.
- Engineered a **LangGraph agentic workflow** over OpenRouter to improve the safety and contextual quality of responses through **prompt-injection detection**, safety routing, sentiment and intent classification, wellness inference, resource discovery, and final response generation.
- Improved trust and continuity in the user experience through structured workflow state, privacy-aware memory controls, and authenticated persistence for conversation history and personalized context.

**AI Delivery Tracking Voice Agent (Vapi, FastAPI, Sanity)** *09/2025*

- Automated delivery status support through a **voice-based AI agent** that handled caller queries end-to-end, translating spoken requests into validated backend lookups and real-time spoken updates.
- Implemented deterministic tool contracts, strict schema validation, and hallucination-safe response generation, improving reliability by restricting outputs to verified backend data and structured fallbacks.
- Added timeout controls, retries, circuit breaking, caching, structured logs, and transcript replay evaluation, strengthening performance, observability, and production readiness for real user interactions.

## Publications

**ORCID ID:**0000-0003-0525-0677 | **Google Scholar**

- "Vision transformer based COVID-19 detection using chest X-rays"
- "SwiftSRGAN–Rethinking Super-Resolution for Efficient and Real-time Inference"
- "Benchmarking Conventional Vision Models on Neuromorphic Fall Detection and Action Recognition Dataset"
- "Efficient Super-Resolution For Chest X-rays"
- "MFAAN: Unveiling Audio Deepfakes with a Multi-Feature Authenticity Network"

*Peer-reviewed work in healthcare ML and computer vision, focused on efficient model design, empirical evaluation, and real-world applicability.*